HDD MASTER TOOL

Quick Reference Sheet — All 16 Tools

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FE Read-only, no changes to disk

WRITES Makes changes to disk

DANGER Irreversible — use with caution

FREE DL Downloads free 3rd-party to

01 Drive Overview — SAFE

Displays all fixed drives with total size, used space, free space, percentage used, volume label, and file system type. Includes a visual fill bar that turns yellow/red when space is low.

Tool: `Get-Volume`

Req: Read-only | No changes made

Tip: Run this first before any maintenance to see current disk state.

02 Health Check (SMART) — SAFE

Reads SMART health status from every physical disk. Shows model, serial number, size, bus type (SATA/NVMe/USB), media type (HDD/SSD), and HealthStatus (Healthy / Warning / Unhealthy).

Tool: `Get-PhysicalDisk`

Req: Read-only | No changes made

Tip: If HealthStatus is NOT 'Healthy' — back up that drive immediately.

03 Disk Cleanup — SAFE

Deletes junk files from: Windows Temp, User Temp, Prefetch, Windows Update download cache, CBS Logs, and empties all Recycle Bins. Reports MB freed after each location.

Tool: `Remove-Item | Clear-RecycleBin`

Req: Administrator required

Tip: Safe to run at any time. Will NOT touch user documents or system files.

04 Drive Optimization — WRITES

Auto-detects drive type and runs the correct operation: TRIM (defrag /L) for SSDs and NVMe, full defragmentation (defrag /U /H) for HDDs. You select the drive letter.

Tool: `defrag.exe (built-in Windows)`

Req: Administrator required | Drive must not be in use

Tip: Never defrag an SSD — the tool handles this automatically.

05 Partition Manager — SAFE

Shows all partitions (letter, size, type, drive number) via Get-Partition. Option to launch the Windows Disk Management GUI (diskmgmt.msc) for visual partition resizing and creation.

Tool: `Get-Partition | diskmgmt.msc`

Req: Administrator for GUI actions

Tip: Use Disk Management GUI to resize/create partitions — safer than CLI.

06 File System Check (CHKDSK) — WRITES

Runs CHKDSK on a selected drive. Choose between: read-only scan (no changes), fix errors (/F), or full scan with bad sector repair (/R). System drive checks are scheduled for next reboot.

Tool: `chkdsk.exe (built-in Windows)`

Req: Administrator required | Drive must not be locked

Tip: Run /R scan on a drive you suspect has physical damage.

07 Disk Usage Analyzer — SAFE

Scans any drive or folder and lists the Top 10 largest files and Top 10 largest folders by size. Helps quickly identify what is consuming the most space.

Tool: `Get-ChildItem (PowerShell built-in)`

Req: Read-only | Scanning takes 10-60 sec for large drives

Tip: Great for finding old backups, ISO files, and video files eating space.

08 Volume Shadow Copies (VSS) — WRITES

Manages Windows Volume Shadow Copy Service snapshots. List all existing shadow copies, create a new snapshot of any drive, or delete a specific shadow copy by ID.

Tool: `vssadmin (built-in Windows)`

Req: Administrator required

Tip: Shadow copies are used by System Restore and Previous Versions. Do NOT delete unless you are sure you don't need restore points.

09 Cloning & Backup — WRITES

Three sub-options: (1) Mirror a folder to another location using Robocopy, (2) Create a full Windows System Image backup via wbadmin, (3) Open Windows Backup & Restore GUI.

Tool: `robocopy | wbadmin | control /name Microsoft.BackupAndRestore`

Req: Administrator required | Destination drive must have enough space

Tip: System image backup requires a separate drive. Cannot back up C: to C:.

10 Disk Imaging — WRITES

Dedicated system image backup using wbadmin. Backs up the entire C: drive (all critical volumes) to a target drive. Also includes shortcuts to Windows Recovery and the Recovery Drive Creator.

Tool: `wbadmin start backup (built-in Windows)`

Req: Administrator required | Target drive cannot be C:

Tip: Run this monthly and store the image on an external drive offsite.

11 Secure Erase — DANGER

Two modes: (1) Free Space Wipe — overwrites only empty space on a drive to remove deleted file traces. (2) Full Drive Wipe — 3-pass NTFS format destroying ALL data. C: drive is blocked.

Tool: `cipher /W | format /p:3`

Req: Administrator required | CAUTION: Full wipe is irreversible

Tip: Use free space wipe for privacy before selling a PC. Full wipe before hardware disposal.

12 Safety Checklist — SAFE

Interactive pre-work checklist before risky operations. Walks through: backup verification, free space check, running processes, Windows update status, and antivirus status. Logs answers.

Tool: `Get-Volume | Get-PhysicalDisk | Get-Process`

Req: Read-only | No changes made

Tip: Always run this checklist before Option 4, 6, 9, 10, or 11.

13 Disk Benchmarking — SAFE

Measures sequential read and write speeds by creating and reading a 512 MB temp file on the selected drive. Reports MB/s speed and deletes the temp file after. No third-party tools needed.

Tool: `PowerShell file I/O (Get-Random + Measure-Command)`

Req: Drive must have 512 MB free | Temp file auto-deleted

Tip: SSD should show 300-3000+ MB/s. HDD typically 80-150 MB/s.

14 Advanced Recovery Tools — FREE DL

Downloads and launches TestDisk and PhotoRec from cgsecurity.org. TestDisk recovers lost partitions and boot records. PhotoRec recovers deleted files (photos, documents, videos) from any drive.

Tool: `TestDisk / PhotoRec (free open-source, downloaded on demand)`

Req: Internet connection for first download | Administrator recommended

Tip: Use TestDisk for partition recovery. Use PhotoRec when files are deleted.

15 Reports & Logs — SAFE

Exports a full drive health report to a text file on the Desktop. Report includes: drive overview, SMART health, partition list, VSS snapshots, and top disk consumers. Timestamped filename.

Tool: `Get-Volume | Get-PhysicalDisk | Get-Partition | Out-File`

Req: Read-only | Report saved to Desktop by default

Tip: Export this report before and after any major disk work as a record.

16 Emergency Triage — SAFE

First-response diagnostic for a failing or suspected drive. Runs all read-only checks in sequence: drive list, SMART status, physical disk detail, partition table, disk errors in Event Log (last 24 hours), and free space status.

Tool: `Get-Volume | Get-PhysicalDisk | Get-Partition | Get-WinEvent`

Req: Administrator required | Read-only, no changes made

Tip: Run this FIRST when a user reports a drive problem. It gives the full picture in one go.

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